Classifying Skin Lesions for Melanoma Detection

Your Role

You are a data scientist consulting with a dermatology research team interested in using image classification to support early skin cancer detection. Skin cancer can be highly treatable when detected early, but manually reviewing thousands of skin lesion images is time consuming and difficult. Your team wants to investigate whether a machine learning model can automate this process by learning how to distinguish benign lesions from malignant ones.

Context

Melanoma is the deadliest skin cancer, accounting for approximately three fourths of all skin cancer deaths. In 2026, the American Cancer Society predicts around 112,000 new cases of melanoma in the United States alone. Early detection is especially important as melanoma outcomes are strongly correlated to how early the cancer is identified and treated.

Your Objective

For this project, you will work with a subset of skin lesion images and metadata from the International Skin Imaging Collaboration dataset. The original dataset has 33,126 images of skin lesions from over 2,000 patients across Australia, Austria, Greece, Spain, and the United States.

You will guide your team through the typical data science workflow: exploring the data, preparing it for modeling, training an image classification model, evaluating its performance, and interpreting the results. By the end, you will create a reproducible analysis that will help your team answer the following research question:

Can an image classification machine learning model distinguish malignant skin lesions from benign ones?